INCH-POUND
MIL-DTL-16377/75A
30 June 2014
SUPERSEDING
MIL-DTL-16377/75(SH)
27 September 2004

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SSL EQUIVALENT, EMERGENCY LIGHTING, 60 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 333.1EM, 333.2EM, 341.1EM, 341.2EM, 342.2EM, 342.3EM, 333.1EMSSL, 333.2EMSSL, 341.1EMSSL, 341.2EMSSL, 342.2EMSSL, AND 342.3EMSSL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

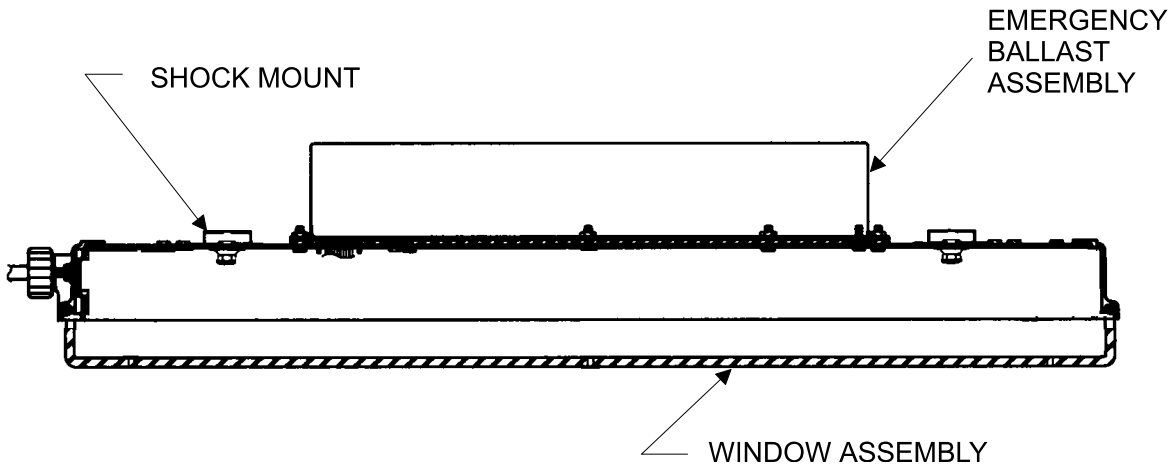
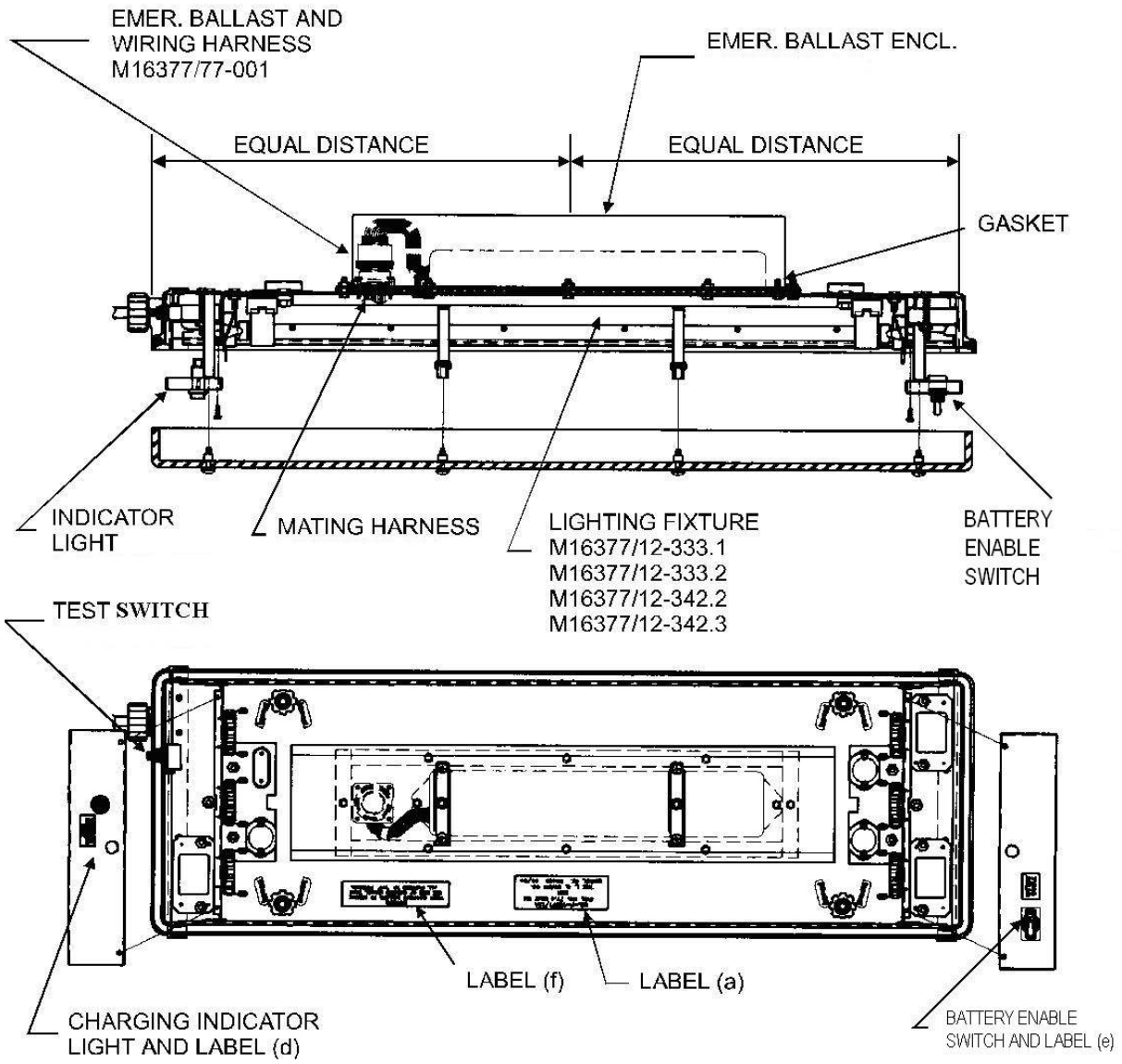


FIGURE 1. Typical outline for symbols 333.1EM, 333.2EM, 341.1EM, 341.2EM, 342.2EM, and 342.3EM.

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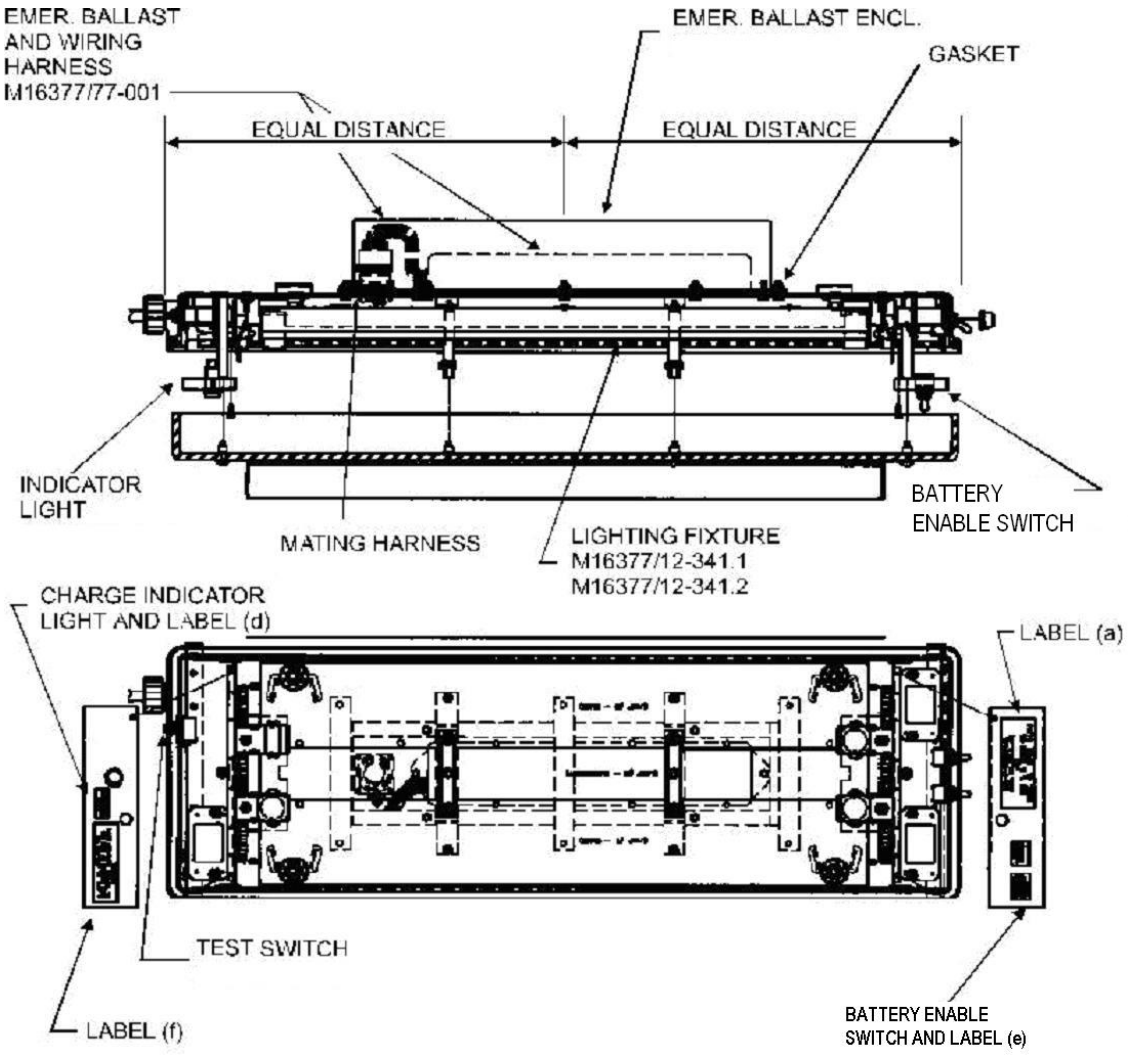


NOTES:

1. For dimensional data, see figures 1, 2, and 3 of MIL-DTL-16377/12.
2. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

FIGURE 2. Typical view of symbols 333.1EM, 333.2EM, 342.2EM, and 342.3EM.

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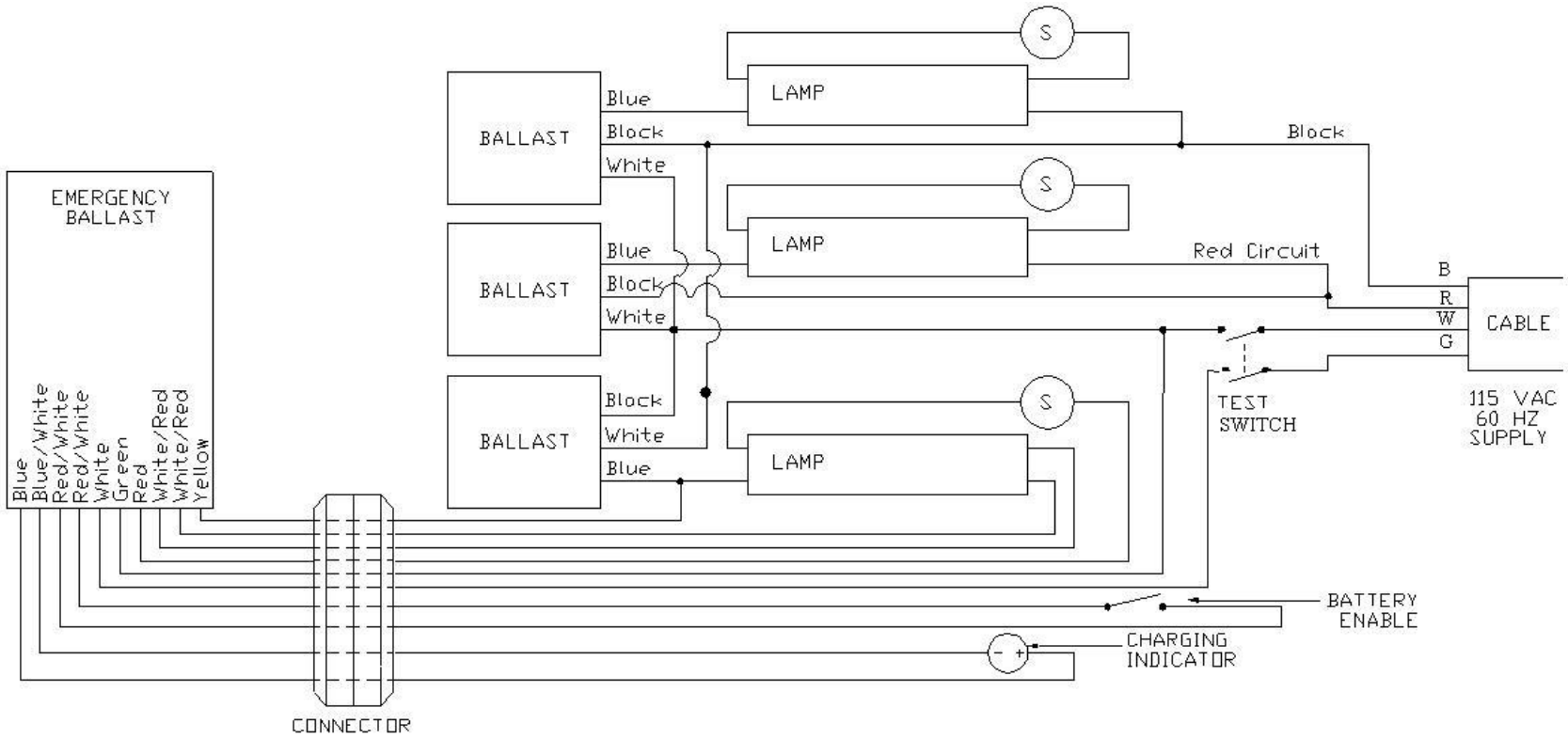


NOTES:

1. For dimensional data, see figures 4 and 5 of MIL-DTL-16377/12.
2. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

FIGURE 3. Typical view of symbols 341.1EM and 341.2EM.

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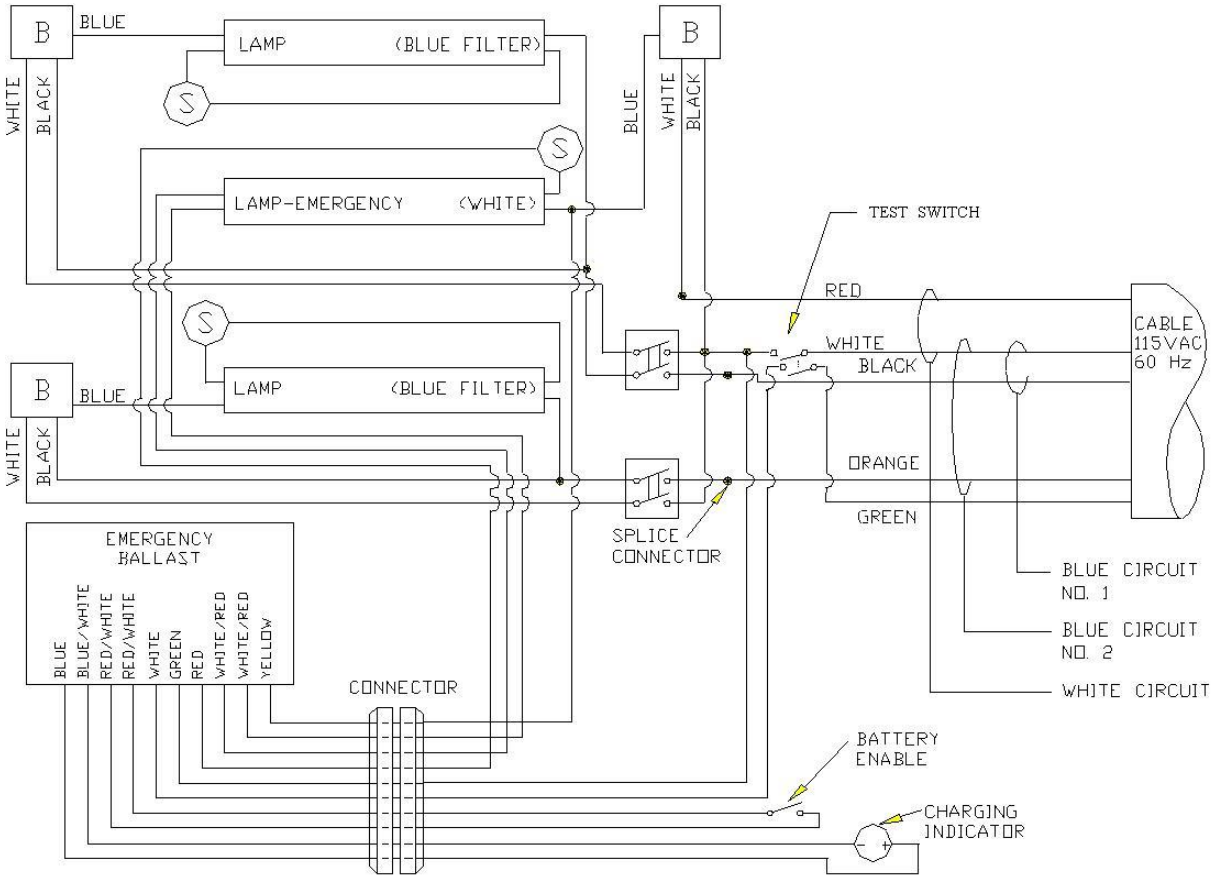


NOTES:

1. Ballast black and white wiring not color coded with input cable black (red) and white, emergency lamp only.
2. Emergency ballast white and green wiring not color coded with input cable white and green.

FIGURE 4. Wiring diagram for symbols 333.1EM, 333.2EM, 342.2EM, and 342.3EM.

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NOTES:

1. Ballast black and white wiring not color coded with input cable black (red) and white, emergency lamp only.
2. Emergency ballast white and green wiring not color coded with input cable white and green.

FIGURE 5. Wiring diagram for symbols 341.1EM and 341.2EM.

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FLUORESCENT REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBERS:

333.1EM: Complete lighting fixture assembly, standard mounting with three 20-watt cool white fluorescent lamps, white diffusing window assembly, and fluorescent emergency ballast.

333.2EM: Complete lighting fixture assembly, flush mounting with three 20-watt cool white fluorescent lamps, white diffusing window assembly, and fluorescent emergency ballast.

341.1EM: Complete lighting fixture assembly, standard mounting, blue and white (Command Information Center [CIC] lighting) clear black shielded with three 20-watt daylight lamps with blue filters over each outside lamp, and fluorescent emergency ballast.

341.2EM: Complete lighting fixture assembly, flush mounting, blue and white CIC lighting clear black shielded with three 20-watt daylight lamps with blue filters over each outside lamp, and fluorescent emergency ballast.

342.2EM: Complete lighting fixture assembly, standard mounting with three 20-watt cool white fluorescent lamps, clear and white prismatic window assembly, and fluorescent emergency ballast.

342.3EM: Complete lighting fixture assembly, flush mounting with three 20-watt cool white fluorescent lamps, clear and white prismatic window assembly, and fluorescent emergency ballast.

DESIGN:

Electrical rating:	60 watts, 120 volts, 60 hertz.
Dimension and configurations:	See figures 1, 2, 3, 4, and 5 of MIL-DTL-16377/12 and figure 10 of MIL-DTL-16377.
Photometric data:	See figures 10 and 11 and tables I and II of MIL-DTL-16377/12.
Lampholder spacing:	See figure 21 of MIL-DTL-16377.
Wiring diagram:	See figures 4 and 5.
Mounting dimensions:	
Symbols 333.1EM, 341.1EM, and 342.2EM:	See figure 25 of MIL-DTL-16377.
Symbols 333.2EM, 341.2EM, and 342.3EM:	See figure 26 of MIL-DTL-16377.
Enclosure:	See 3.5.11 of MIL-DTL-16377.
Symbols 333.1EM, 333.2EM, 342.2EM, and 342.3EM	Watertight.
Symbols 341.1EM and 341.2EM	Dripproof 15 degrees.
Cable:	
Symbols 333.1EM, 333.2EM, 342.2EM, and 342.3EM:	10 feet of Type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Symbols 341.1EM and 341.2EM:	10 feet of Type LSMHOF-7 in accordance with MIL-DTL-24643/7.
Gasket:	Neoprene rubber (durometer 35-45).
Weight:	
Symbols 333.1EM and 342.2EM:	20 pounds maximum.
Symbols 333.2EM and 342.3EM:	22 pounds maximum.
Symbol 341.1EM:	23 pounds maximum.
Symbol 341.2EM:	25 pounds maximum.

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Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377. Labels (a), (b), and (c) in accordance with MIL-DTL-16377/12. Labels (d), (e), and (f) as shown on figures 2 and 3:

a. Two Type I labels.
b. One Type II label for Symbols 333.2EM and 342.2EM only.
c. One Type IV label. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table I by name and stock number only.
d. One Type IV label. The information to be shown on the Type IV label shall be "CHARGING INDICATOR LIGHT."
e. One Type IV label. The information to be shown on the Type IV label shall be "BATTERY ENABLE."
f. One Type V label. The information to be shown on the Type V label shall be "DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION."

Standard Parts: The standard parts and their respective part numbers shown in table I shall be used in the construction of the fixtures specified herein.

SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBERS:

333.1EMSSL: Complete lighting fixture assembly, standard mounting with LED light source, white diffusing window assembly, and emergency battery back-up.
333.2EMSSL: Complete lighting fixture assembly, flush mounting with LED light source, white diffusing window assembly, and emergency battery back-up.
341.1EMSSL: Complete lighting fixture assembly, standard mounting, blue and white CIC lighting clear black shielded with white and blue LED light sources, and emergency battery back-up.
341.2EMSSL: Complete lighting fixture assembly, flush mounting, blue and white CIC lighting clear black shielded with white and blue LED light sources, and emergency battery back-up.
342.2EMSSL: Complete lighting fixture assembly, standard mounting with LED light source, clear and white prismatic window assembly, and emergency battery back-up.
342.3EMSSL: Complete lighting fixture assembly, flush mounting with LED light source, clear and white prismatic window assembly, and emergency battery back-up.

DESIGN:

Electrical rating: 60 watts maximum, 120 volts, 60 hertz.
Dimension and configurations: External dimensions shall not exceed those shown on figures 1, 2, 3, 4, and 5 of MIL-DTL-16377/12 and figure 10 of MIL-DTL-16377.
Photometric data: See figures 10 and 11 and tables I and II of MIL-DTL-16377/12 and 3.7.1 of MIL-DTL-16377.
Emergency mode lumen output: Luminaire shall provide a minimum of 240 lumens on a fully charged battery and maintain over 144 lumens over a 5-hour runtime.
Chromaticity: See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit: See 3.17 of MIL-DTL-16377.
Wiring: Shall maintain same control scheme as legacy fluorescent, including battery enable switch, battery charging indicator.

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Mounting dimensions:

Symbols 333.1EMSSL, 341.1EMSSL, and 342.2EMSSL:	See figure 25 of MIL-DTL-16377.
Symbols 333.2EMSSL, 341.2EMSSL, and 342.3EMSSL:	See figure 26 of MIL-DTL-16377.

Enclosure: See 3.5.11 of MIL-DTL-16377.

Symbols 333.1EM, 333.2EM, 342.2EM, and 342.3EM	Watertight.
Symbols 341.1EM and 341.2EM	Dripproof 15 degrees.

Cable:

Symbols 333.1EMSSL, 333.2EMSSL, 342.2EMSSL, and 342.3EMSSL:	10 feet of Type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Symbols 341.1EMSSL and 341.2EMSSL:	10 feet of Type LSMHOF-7 in accordance with MIL-DTL-24643/7.

Gasket: Neoprene rubber (durometer 35-45), if design uses legacy fluorescent ballast enclosure.

Weight:

Symbols 333.1EMSSL and 342.2EMSSL:	20 pounds maximum.
Symbols 333.2EMSSL and 342.3EMSSL:	22 pounds maximum.
Symbol 341.1EMSSL:	23 pounds maximum.
Symbol 341.2EMSSL:	25 pounds maximum.

Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377. Labels (a), (b), and (c) in accordance with MIL-DTL-16377/12. Labels (d), (e), and (f) as shown on figures 2 and 3:

a. Two Type I labels.

b. One Type II label for Symbols 333.2EM and 342.2EM only.

c. One Type IV label. The information to be shown on the Type IV label shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by:

(1) A list of all standard parts shown in table I by name and stock number only, if appropriate, or

(2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer’s part numbers.

d. One Type IV label. The information to be shown on the Type IV label shall be “CHARGING INDICATOR LIGHT.”

e. One Type IV label. The information to be shown on the Type IV label shall be “BATTERY ENABLE.”

f. One Type V label. The information to be shown on the Type V label shall be “DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION.”

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BATTERY BACK-UP SYSTEM REQUIREMENTS:

Battery back-up system for Type III luminaires.

DESIGN:

Assembly:	The battery back-up system consists of a high temperature, rechargeable battery; battery charger; AC power sensing circuit; wiring harness; enclosure.
Circuit design:	Back-up system shall incorporate Thyristor devices for protection against high voltage spikes.
Battery back-up location:	Battery back-up assembly shall fit within the dimensions of figure 1 of MIL-DTL-16377/8 or within the legacy fluorescent emergency ballast enclosure, as described in table I.
Battery material:	Use of other than NI-CAD battery systems requires specific authorization from NAVSEA.
Emergency operation time:	5 hours minimum.
Full recharge time:	20 hours maximum.
Temperature rating:	131 °F (55 °C) minimum.
Charging power:	4.0 watts maximum.
System power factor:	>0.95.
System harmonics:	<3 percent.
Wiring:	Connection between battery back-up system and the rest of the luminaire shall be made with a commercial wiring harness that allows for easy replacement of battery back-up system.

EXAMINATION AND TESTS:

Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II for Type I fixtures and table III for Type III luminaires.
Conformance:	
Comparison inspection:	Same as first article inspection.
Conformance inspection:	See MIL-DTL-16377.

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TABLE I. Standard parts.

Name	Military P/N or Specification	National Stock Number (NSN)	Remarks
Complete fixture, Symbol 333.1	M16377/12-333.1	6210-01-465-6342	For use with Symbol 333.1EM
Complete fixture, Symbol 333.2	M16377/12-333.2	6210-01-135-2016	For use with Symbol 333.2EM
Complete fixture, Symbol 341.1	M16377/12-341.1	6210-00-455-0068	For use with Symbol 341.1EM
Complete fixture, Symbol 341.2	M16377/12-341.2	6210-01-107-2581	For use with Symbol 341.2EM
Complete fixture, Symbol 342.2	M16377/12-342.2	6210-01-133-6655	For use with Symbol 342.2EM
Complete fixture, Symbol 342.3	M16377/12-342.3	6210-01-283-0375	For use with Symbol 342.3EM
Window assembly, white diffusing	M16377/12-001	6210-00-295-2849	
Window assembly, clear black shielded	M16377/12-003	6210-00-725-9030	
Window assembly, clear and white prismatic	M16377/12-004	6210-00-814-5914	
Mating harness	AMP part No. 206043-1 or equal		Mates with M16377/77-001
Emergency ballast and wiring harness	M16377/77-001	6250-01-491-7039	
Indicator light	Commercial	6220-01-415-7889	LED type
Battery enable switch [1/]	Commercial	5930-01-414-9240	DPST
Enclosure for emergency ballast			Aluminum; approximately 3″ H x 3 ½″ W x 16″ L
Test switch, push	Commercial	5930-01-416-0342	Double pole, test switch with rubber boot
Switch cover, rubber boot	MIL-DTL-5423/2		Neoprene
Red filter tube	M16377/42-003	6210-00-295-2838	Not furnished, see 3.7.9 of MIL-DTL-16377
Broad band blue filter tube	M16377/42-005	6210-00-441-5138	Not furnished, see 3.7.9 of MIL-DTL-16377
Amber filter tube	M16377/42-006	6210-01-166-7184	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (1% transmission)	M16377/42-007	6210-01-498-8001	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (2.5% transmission)	M16377/42-009	6210-01-313-8168	Not furnished, see 3.7.9 of MIL-DTL-16377

NOTE:

[1/] Switch shall normally be left in the “Battery Enable” position.

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TABLE II. First article and comparison inspection for complete type I fixtures. 1/

Tests	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	
Light output	3.7.1	4.8.6.1	Candlepower
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Emergency light output	3.7.1.1	4.8.6.8	Lumen output
Emergency battery back-up performance: Operating time and discharge test	3.6.19.3	4.8.26	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight for Symbols 333.1EM, 333.2EM, 342.2EM, and 342.3EM.
Dripproof 15 degrees for Symbols 341.1EM and 341.2EM.

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TABLE III. First article and comparison inspection for complete type III luminaires. [1/]

Tests	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Enclosure effectiveness	3.5.11	4.8.14	[2/, 3/]
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14	[2/, 3/]
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.14	4.8.3	[2/]
Noise	3.5.14	4.8.9	[2/]
Continuity of grounding	3.6.13.4	4.8.12	
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT)	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Emergency light output	3.7.1.1	4.8.6.8	Lumen output
Emergency battery back-up performance: Operating time and discharge test	3.6.19.3	4.8.26	

NOTES:

[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

[2/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

[3/] Watertight for Symbols 333.1EMSSL, 333.2EMSSL, 342.2EMSSL, and 342.3EMSSL. Dripproof 15 degrees for Symbols 341.1EMSSL and 341.2EMSSL.

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TABLE IV. Part number and general information.

Military Part Number M16377/75-	Item Name	National Stock Number (NSN)
333.1EM	Complete fixture, Symbol 333.1EM	6210-01-478-5205
333.2EM	Complete fixture, Symbol 333.2EM	6210-01-560-3546
341.1EM	Complete fixture, Symbol 341.1EM	6210-01-569-0601
341.2EM	Complete fixture, Symbol 341.2EM	
342.2EM	Complete fixture, Symbol 342.2EM	6210-01-568-8510
342.3EM	Complete fixture, Symbol 342.3EM	
333.1EMSSL	Complete fixture, solid state, Symbol 333.1EMSSL	
333.2EMSSL	Complete fixture, solid state, Symbol 333.2EMSSL	
341.1EMSSL	Complete fixture, solid state, Symbol 341.1EMSSL	
341.2EMSSL	Complete fixture, solid state, Symbol 341.2EMSSL	
342.2EMSSL	Complete fixture, solid state, Symbol 342.2EMSSL	
342.3EMSSL	Complete fixture, solid state, Symbol 342.3EMSSL	

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy – SH
DLA – IS

Preparing activity:
Navy – SH
(Project 6210-2012-035)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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